

Question 1 (20 points). Let $\alpha \in \mathbb{R}$ be a constant and consider the following processes

$$X_t = W_t \cos(\alpha t) + W_{t-1} \sin(\alpha t),$$

where $\{W_t\} \stackrel{iid}{\sim} N(0, \sigma_w^2)$. Find the *mean*, *variance*, *autocovariance function* and *autocorrelation function*. Is $\{X_t\}$ a weakly stationary process?

$$\begin{aligned} E[X_t] &= E[W_t \cos(\alpha t) + W_{t-1} \sin(\alpha t)] = \cos(\alpha t) E[W_t] + E[W_{t-1}] \sin(\alpha t) \\ &= 0 + 0 = 0 \end{aligned} \quad (4)$$

$$\gamma_{t+h} = \text{cov}(X_{t+h}, X_t)$$

$$= E\left[\left[W_{t+h} \cos(\alpha(t+h)) + W_{t-1} \sin(\alpha(t+h))\right] \left[W_t \cos(\alpha t) + W_{t-1} \sin(\alpha t)\right]\right]$$

$$= E[W_{t+h} W_t] \cos(\alpha(t+h)) \cos(\alpha t)$$

$$+ E[W_{t+h} W_{t-1}] \cos(\alpha(t+h)) \sin(\alpha t)$$

$$+ E[W_{t-1} W_t] \sin(\alpha(t+h)) \cos(\alpha t)$$

$$+ E[W_{t-1} W_{t-1}] \sin(\alpha(t+h)) \sin(\alpha t)$$

(3)

Question 1 (Continued).

$$h=0 \rightarrow \gamma_{t+h,t} = \sigma_w^2$$

$$h=-1 \rightarrow \gamma_{t+h,t} = \sigma_w^2 \cos(\alpha(t-1)) \sin(\alpha t) \Rightarrow \gamma_{t+h,t} = \begin{cases} \sigma_w^2 & h=0 \\ \sigma_w^2 \cos(\alpha(t-1)) \sin(\alpha t) & h=-1 \\ \sigma_w^2 \cos(\alpha t) \sin(\alpha(t+1)) & h=+1 \end{cases}$$

$$h=+1 \rightarrow \gamma_{t+h,t} = \sigma_w^2 \cos(\alpha t) \sin(\alpha(t+1))$$

$$|h| > 2 \rightarrow \gamma_{t+h,t} = 0$$

(5)

The variance: $\text{Var}(X_t) = \gamma_{t,t} = \sigma_w^2$ (2)

$$\rho_{t+h,t} = \frac{\gamma_{t+h,t}}{\text{Var}(X_t)} = \begin{cases} 1 & h=0 \\ \cos(\alpha(t-1)) \sin(\alpha t) & h=-1 \\ \cos(\alpha t) \sin(\alpha(t+1)) & h=+1 \end{cases}$$
 (3)

X_t isn't weakly stationary in general

but if $\alpha = k\pi$ $k \in \mathbb{Z}$ then it's stationary (3)

Question 2 (20 points). Consider the following mixed ARMA(2,2) model

$$X_t = -\frac{1}{5}X_{t-1} + \frac{12}{25}X_{t-2} + W_t + \frac{3}{5}W_{t-1} - \frac{4}{25}W_{t-2},$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$.

- Determine the correct orders p and q of the ARMA(p, q) model, if necessary. Briefly discuss why over-parametrized models should be avoided in general.
- This model is both causal and invertible. Why?
- Compute the coefficients ψ_j of the causal representation of this time series,

$$X_t = W_t + \sum_{j=1}^{\infty} \psi_j W_{t-j}.$$

- Compute the coefficients π_j of the invertible representation of this time series,

$$W_t = X_t + \sum_{j=1}^{\infty} \pi_j X_{t-j}.$$

- Is the behaviour of the ψ - and π -weights what you expect, given the type of model? Explain.

$$\begin{aligned} a) \quad & \left(1 + \frac{1}{5}B - \frac{12}{25}B^2\right) X_t = \left(1 + \frac{3}{5}B - \frac{4}{25}B^2\right) W_t \\ & \left(1 - \frac{3}{5}B\right) \left(1 + \frac{4}{5}B\right) X_t = \left(1 - \frac{1}{5}B\right) \left(1 + \frac{4}{5}B\right) W_t \\ & \left(1 - \frac{3}{5}B\right) X_t = \left(1 - \frac{1}{5}B\right) W_t \end{aligned}$$

4 or finding roots

$\Rightarrow$ ARMA (1,1) process with $X_t = \frac{3}{5}X_{t-1} + W_t - \frac{1}{5}W_{t-1}$
it was over parametrized

2

Question 2 (Continued).

$$b) \phi(B) = 1 - \frac{3}{5}B \Rightarrow B = \frac{5}{3} \rightarrow |B| > 1 \rightarrow \text{causal} \quad (2)$$

$$\theta(B) = 1 - \frac{1}{5}B \Rightarrow B = 5 \rightarrow |B| > 1 \rightarrow \text{invertible} \quad (2)$$

$$c) \psi(B) = \frac{\theta(B)}{\phi(B)} = \frac{1 - \frac{1}{5}B}{1 - \frac{3}{5}B}$$

$$= (1 - \frac{1}{5}B) (1 + \frac{3}{5}B + \frac{9}{25}B^2 + \dots) \quad (2)$$

$$= 1 + \frac{2}{5}B + \frac{6}{25}B^2 + \dots$$

$$= 1 + \sum_{j=1}^{\infty} 2 \left(\frac{3^{j-1}}{5^j} \right) B^j$$

$$\psi_0 = 1, \quad \psi_j = 2 \left(\frac{3^{j-1}}{5^j} \right) \quad (2)$$

Question 2 (Continued).

$$d) \pi(B) = \frac{\phi(B)}{\theta(B)} = \frac{1 - \frac{3}{5}B}{1 - \frac{1}{5}B}$$

$$= (1 - \frac{3}{5}B) \left(1 + \frac{B}{5} + \frac{B^2}{25} + \dots \right) \quad (2)$$

$$= 1 - \frac{2}{5}B - \frac{2}{25}B^2 - \frac{2}{125}B^3 + \dots$$

$$= 1 + \sum_{j=1}^{\infty} (-2) \left(\frac{1}{5} \right)^j B^j$$

$$\pi_0 = 1, \pi_j = (-2) \left(\frac{1}{5} \right)^j \quad j = 1, 2, \dots$$

(2)

e) since the weights decay exponentially, the process is stationary & it's also invertible & causal

(2)

Question 3 (20 points). Suppose we are given 1500 observations $x_1, x_2, \dots, x_{1500}$ of a particular time series, which are used to compute the sample autocovariance function at different lag values as follows:

$$\hat{\gamma}_0 = 5, \quad \hat{\gamma}_1 = 0, \quad \hat{\gamma}_2 = 2.5, \quad \hat{\gamma}_3 = 0, \quad \hat{\gamma}_4 = 1.$$

Estimate the coefficients ϕ_1, ϕ_2 , and the noise variance σ_w^2 by deriving and solving the Yule-Walker equations for the following AR(2) model:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t,$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$.

$$\gamma_h = \phi_1 \gamma_{h-1} - \phi_2 \gamma_{h-2} + \begin{cases} \sigma_w^2 & h=0 \\ 0 & h \neq 0 \end{cases} \quad (2)$$

$$\begin{cases} h=0 \rightarrow \gamma_0 - \phi_1 \gamma_1 - \phi_2 \gamma_2 = \sigma_w^2 \rightarrow \hat{\sigma}_w^2 = \gamma_0 - \hat{\phi}_1 \gamma_1 - \hat{\phi}_2 \gamma_2 = 3.75 \quad (5) \\ h=3 \rightarrow \gamma_3 - \phi_1 \gamma_2 - \phi_2 \gamma_1 = 0 \rightarrow \hat{\gamma}_3 - \hat{\phi}_1 \hat{\gamma}_2 - \hat{\phi}_2 \hat{\gamma}_1 = 0 \quad (5) \\ \Rightarrow \hat{\phi}_1 = 0 \\ h=2 \rightarrow \gamma_2 - \phi_1 \gamma_1 - \phi_2 \gamma_0 = 0 \rightarrow \hat{\phi}_2 = 0.5 \quad (5) \end{cases}$$